Unit 2 - Linear Algebra
2.5 Determinants and Inverses

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MSDS 520: Math and Stats Foundations for Data Science
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Outline

1. Why Determinants and Inverses?
2. The Determinant as Area
3. When the Determinant Is Zero
4. The Inverse
5. Solving Systems with the Inverse
6. A First Look at Eigenvectors
7. Review

Why Determinants and Inverses?

A Problem to Motivate Us

Back to the pharmacy (Unit 2-1)

A 10 mg/mL stock and a 2 mg/mL stock. Tonight's order: 500 mL at 4 mg/mL. As one matrix equation:

$$\begin{bmatrix} 1 & 1 \\ 10 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 500 \\ 2000 \end{bmatrix}$$

- Row 1: total volume, mL.
- Row 2: total drug, mg.

But the orders keep coming

order	**mL**	**mg/mL**	**mg**
1	500	4	2000
2	300	6	1800
3	1000	3	3000

Same left side every time. Only the right-hand side changes.

Two questions this lecture answers

1. Will **every** order have exactly one answer? $\Rightarrow$ the **determinant**
2. Can we do the work **once** and reuse it for every order? $\Rightarrow$ the **inverse**

Where This Shows Up in Data Science

Determinants and inverses underneath the tools

- **Fitting a regression.** The best-fit coefficients are $(X^\top X)^{-1} X^\top \bar{y}$: an inverse. Unit 3 builds this.
- **Redundant features.** When two columns of a data table carry the same information (**collinearity**), a determinant goes to 0 and the coefficients become unstable.
- **Unusual lab results.** The Mahalanobis distance, which flags a patient whose values are odd **in combination**, uses the inverse of a covariance matrix.
- **The bell curve in many dimensions.** The multivariate normal's formula contains both $\det \Sigma$ and Σ^{-1}.
- **Eigenvectors.** Google's PageRank and PCA, a standard way to compress data, are both built on them. Today's last section introduces them.

What We Will Build

The plan

1. The **determinant** as an **area scale factor**: the formula $ad - bc$ and where it comes from; **volume** in 3D.
2. Its **sign** (orientation), and its link to the **dot product**.
3. When $\det = 0$: collapse, dependent columns, and Unit 2-1's three outcomes.
4. The **inverse**: undoing a transformation, the 2×2 formula, and the rules.
5. **Solving** $A\bar{x} = \bar{b}$ as $\bar{x} = A^{-1}\bar{b}$, and Python.
6. A first look at **eigenvectors**: the directions that do not turn, found with $\det(A - \lambda I) = 0$.

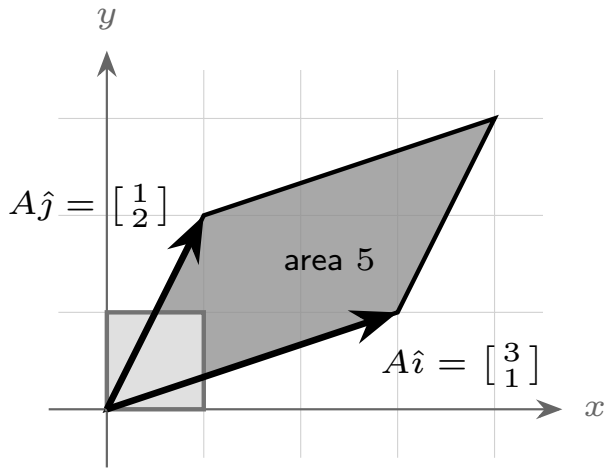
Notebooks for this lecture: `U2-5_Determinants-1` and `-2`; for the eigen section, `U2-6_Eigen-1` and `-2`.

The Determinant as Area

How Much Does a Matrix Stretch Area?

Recall from Unit 2-4

- The columns of A are where $\hat{\imath}$ and $\hat{\jmath}$ land.
- So the **unit square** (area 1) lands on the **parallelogram** the two columns span.
- **Question:** what is that parallelogram's area?



Definition: the determinant

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$$

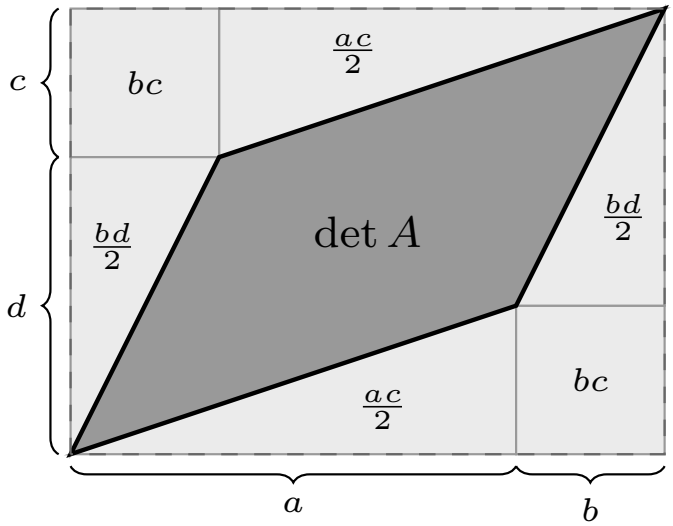
It is the (signed) factor by which A scales **any** area. The unit square is just the easiest example.

Worked: $A = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$

$\det A = 3 \cdot 2 - 1 \cdot 1 = 5$.
Every area is multiplied by 5.

Notebook `U2-5_Determinants-1_AsArea`, §1

Why $ad - bc$? The Box Minus the Corners



Drawn for $A = \left[\begin{smallmatrix} 3 & 1 \\ 1 & 2 \end{smallmatrix}\right]$.

Area of the parallelogram

- The enclosing box is $(a + b)$ wide and $(c + d)$ tall.
- Cut off two triangles of $\frac{ac}{2}$, two of $\frac{bd}{2}$, and two rectangles of bc.

$$\begin{aligned} \text{area} &= (a+b)(c+d) - ac - bd - 2bc \\ &= ac + ad + bc + bd - ac - bd - 2bc \\ &= ad - bc \end{aligned}$$

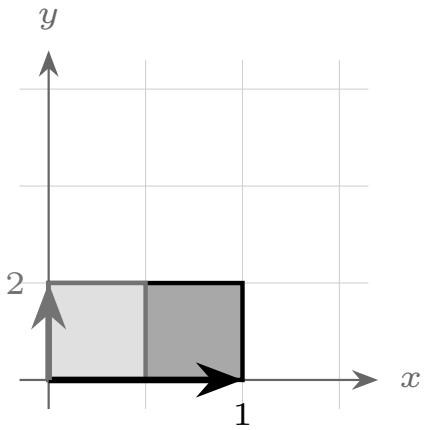
Check with $A = \left[\begin{smallmatrix} 3 & 1 \\ 1 & 2 \end{smallmatrix}\right]$

Box $4 \times 3 = 12$. Corners $3 + 2 + 2 = 7$.
Area $12 - 7 = 5 = \det A$.

The picture has every entry positive, but the formula $ad - bc$ holds for every 2×2 matrix.

Parallelograms and Their Determinants

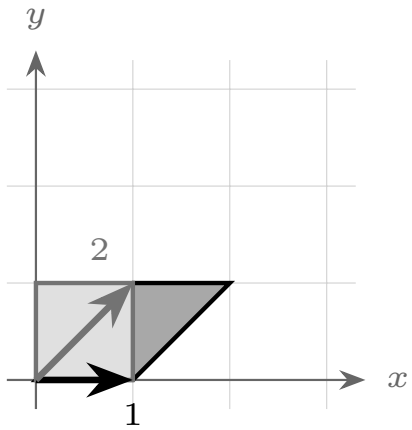
The unit square and its image under four matrices, all drawn to the same scale. Each arrow is labelled with its column number.



$$\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

$\det = 2 - 0 = 2$

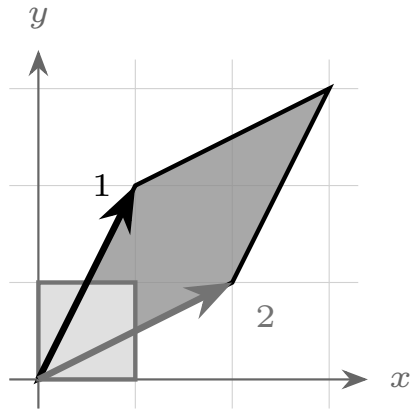
Area **doubles**.



$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$\det = 1 - 0 = 1$

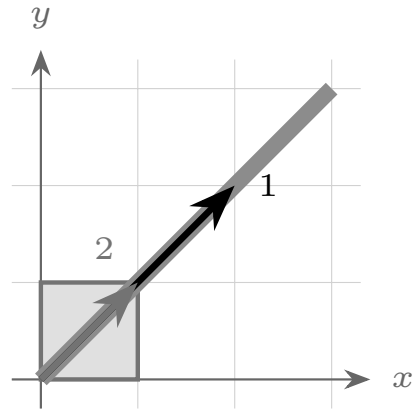
Tilted: area **kept**.



$$\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

$\det = 1 - 4 = -3$

Area $\times 3$, **flipped**.



$$\begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}$$

$\det = 2 - 2 = 0$

Flat: area 0.

Size of $\det$**:** the area. **Sign:** the orientation (next slides).

Determinants of the Transformations You Know

Transformation	**Matrix**	$\det$	**What happens to area**
Scaling	$\begin{bmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$	$2 \cdot \frac{1}{2} = 1$	kept
Uniform scaling	$\begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}$	k^2	multiplied by k^2
Shear	$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$	1	kept: same base and height
Rotation	$R(\theta)$	$\cos^2\theta + \sin^2\theta = 1$	kept
Reflection	$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$	-1	kept, but **flipped**
Projection	$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$	0	crushed to 0

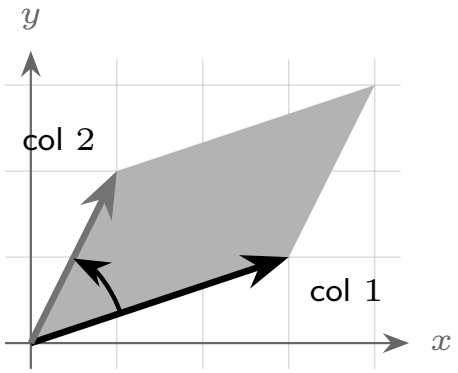
Two moves in a row: the factors multiply

$$\det(AB) = \det A \cdot \det B$$

- B multiplies area by $\det B$, then A multiplies **that** by $\det A$.
- **Example:** rotate, then double everything. $\det = 1 \cdot 4 = 4$. Check: $\det \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix} = 0 + 4 = 4$.

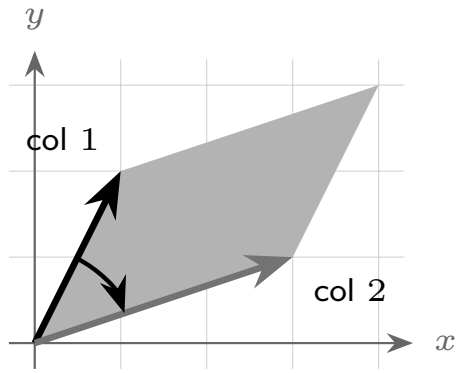
The Sign: Orientation

Same two columns, opposite order. Same parallelogram, same area 5: only the **sign** changes.



$$\det \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} = 6 - 1 = +\mathbf{5}$$

Counterclockwise



$$\det \begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix} = 1 - 6 = -\mathbf{5}$$

Clockwise

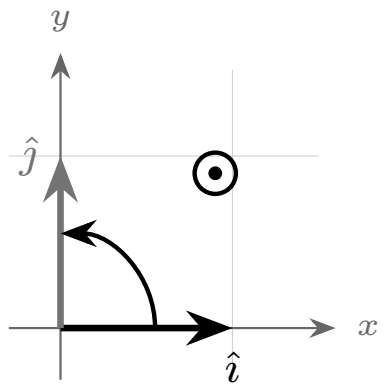
In general: swapping the columns flips the sign

$$\det \begin{bmatrix} b & a \\ d & c \end{bmatrix} = bc - ad = -(ad - bc) = -\det \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

The Sign: the Right-Hand Rule

The rule

1. Point your right hand's fingers along **column** 1.
2. Curl them toward **column** 2, the short way round.
3. Thumb **out of** the screen ($\odot$): $\det > 0$. Thumb **into** it ($\otimes$): $\det < 0$, **mirrored**.



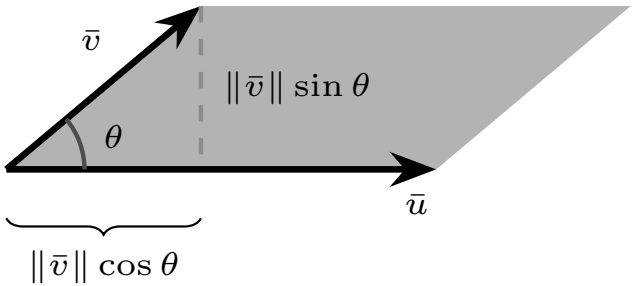
$\hat{\imath}$ to $\hat{\jmath}$: thumb **out**, along the z-axis.

Check it on matrices you know

- $I = \left[\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right]$: columns $\hat{\imath}$ then $\hat{\jmath}$, thumb out, $\det I = +1$.
- Reflection $\left[\begin{smallmatrix} -1 & 0 \\ 0 & 1 \end{smallmatrix}\right]$: from $\left[\begin{smallmatrix} -1 \\ 0 \end{smallmatrix}\right]$ to $\left[\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right]$ is clockwise, thumb in, $\det = -1$.
- In 3D the thumb **is** the third axis: $\det > 0$ when column 3 sits on the thumb's side.

Notebook `U2-5_Determinants-1_AsArea`, §3, the sign

The Determinant and the Dot Product



Shadow along $\bar{u}$: the cosine. Height off $\bar{u}$: the sine.

Dot product: the **parallel** part

$$\bar{u} \cdot \bar{v} = \|\bar{u}\| \, \|\bar{v}\| \cos\theta$$

How much of $\bar{v}$ lies **along** $\bar{u}$. Zero when they are **perpendicular**.

Determinant: the **perpendicular** part

$$\det \begin{bmatrix} \bar{u} & \bar{v} \end{bmatrix} = \|\bar{u}\| \, \|\bar{v}\| \sin\theta$$

Base $\|\bar{u}\|$ times height $\|\bar{v}\| \sin\theta$. Zero when they are **parallel**.

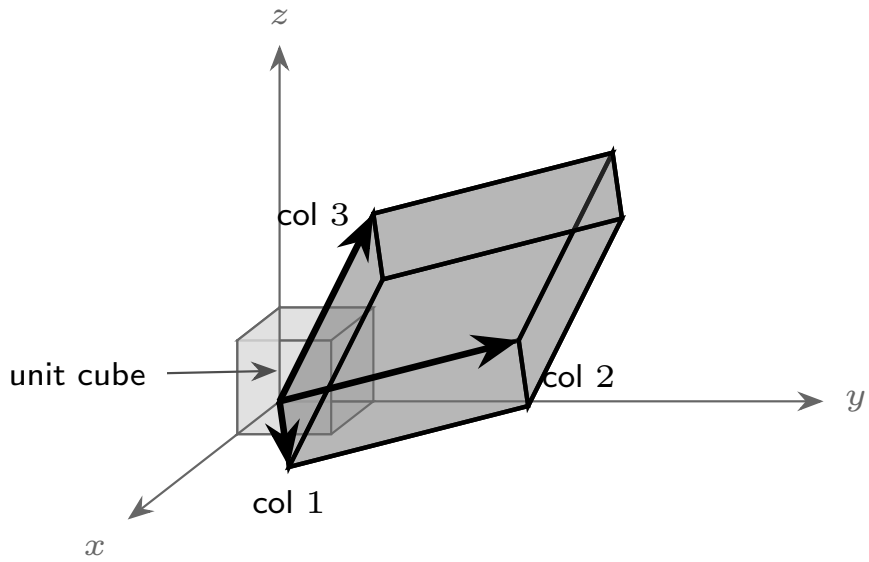
So: $\det = 0$ exactly when the columns are parallel

Parallel means one column is a multiple of the other: the columns are **linearly dependent** (Unit 2-2). (Here θ is measured counterclockwise from $\bar{u}$ to $\bar{v}$, which is what lets the sine carry the sign.)

Bigger Matrices: Volume and Beyond

Same meaning, more dimensions

- 2×2: the **area** factor.
- 3×3: the **volume** factor. The unit cube becomes a slanted box, a **parallelepiped**.
- $n \times n$: the **hyper-volume** factor.
- Sign: orientation. $\det = 0$: flattened.



The box spanned by the columns of $A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix}$.

But the formula gets cumbersome

- Terms in the formula: 2 for 2×2, 6 for 3×3, 24 for 4×4, over 3.6 million for 10×10.
- So computers find $\det$ by **elimination** instead.

Exercise: Determinants

Try it

1. Compute each determinant and say what the matrix does to area:

$$\begin{bmatrix} 4 & 0 \\ 0 & 3 \end{bmatrix}, \qquad \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix}, \qquad \begin{bmatrix} 0 & 2 \\ 3 & 0 \end{bmatrix}, \qquad \begin{bmatrix} 2 & -1 \\ 4 & -2 \end{bmatrix}.$$

2. A triangle of area 6 is transformed by $\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$. What is the area of its image?

3. For which values of k does $\begin{bmatrix} k & 4 \\ 1 & k \end{bmatrix}$ squash the plane flat?

Exercise: Solution

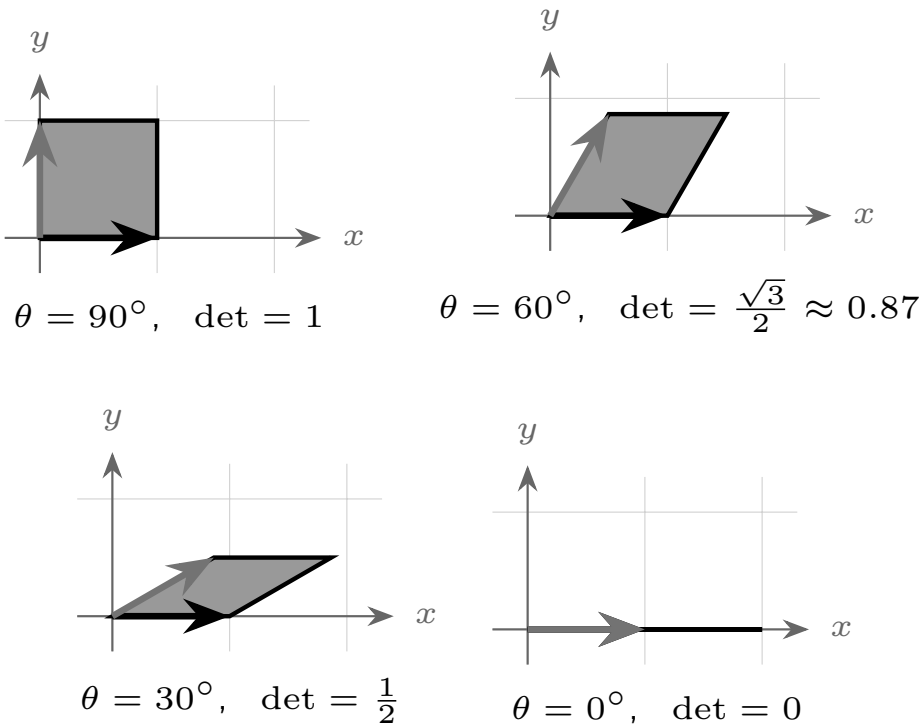
Answers

1. The four area factors:
   - $4 \cdot 3 - 0 \cdot 0 = 12$: x is stretched by 4 and y by 3, so area is multiplied by 12.
   - $2 \cdot 3 - 5 \cdot 1 = 1$: the shape is distorted, but its area is **kept**.
   - $0 \cdot 0 - 2 \cdot 3 = -6$: area times 6, and **flipped**. Here $\hat{\imath} \mapsto \left[\begin{smallmatrix} 0 \\ 3 \end{smallmatrix}\right]$ and $\hat{\jmath} \mapsto \left[\begin{smallmatrix} 2 \\ 0 \end{smallmatrix}\right]$: the columns now turn clockwise.
   - $2(-2) - (-1)(4) = -4 + 4 = 0$: area **crushed to** 0. Column 2 is $-\frac{1}{2}$ times column 1.
2. Every area is multiplied by $\det = 3 \cdot 2 - 1 \cdot 1 = 5$, so the image has area $6 \times 5 = 30$. The triangle's shape does not matter.
3. Flat means area 0: $\det = k \cdot k - 4 \cdot 1 = k^2 - 4 = 0$, so $k = 2$ or $k = -2$. Check $k = 2$: $\left[\begin{smallmatrix} 2 & 4 \\ 1 & 2 \end{smallmatrix}\right]$ has column $2 = 2\times$ column 1.

When the Determinant Is Zero

Watching the Area Drain Away

Hold column 1 at $\left[\begin{smallmatrix}1\\0\end{smallmatrix}\right]$ and swing column 2, also a unit vector, down toward it: $A(\theta) = \left[\begin{smallmatrix}1 & \cos\theta\\ 0 & \sin\theta\end{smallmatrix}\right]$, so $\det A(\theta) = \sin\theta$.



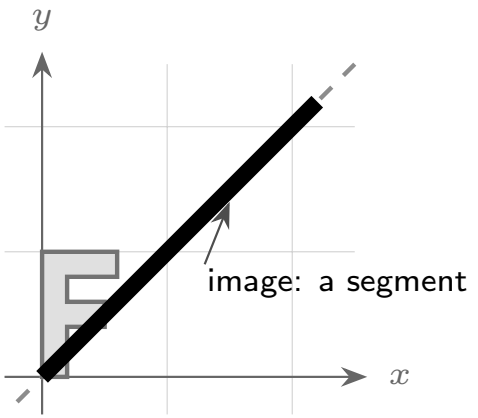
- As the columns line up, the parallelogram **flattens** and its area drains to 0.
- At $\theta = 0°$ the whole plane lands on the x-axis: a **collapse** to a line.

Notebook `U2-5_Determinants-1_AsArea`, §2

$\det = 0$: Three Ways of Saying One Thing

$$M = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}$$

$$\det M = 2 \cdot 1 - 1 \cdot 2 = 0$$



M lands the whole F on the line $y = x$.

One fact, three statements

1. **Geometric: zero area.** M squashes the whole plane onto one line, $y = x$.
2. **Redundancy: the columns are linearly dependent** (Unit 2-2). Column $1 = \left[\begin{smallmatrix} 2 \\ 2 \end{smallmatrix}\right]$ is $2\times$ column $2 = \left[\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right]$.
3. **Systems: never exactly one solution** (Unit 2-1). Every $M\bar{x}$ lies on $y = x$, so $M\bar{x} = \bar{b}$ has
   - **no** solutions if $\bar{b}$ is off the line ($\left[\begin{smallmatrix} 3 \\ 5 \end{smallmatrix}\right]$);
   - **infinitely many** if $\bar{b}$ is on it ($\left[\begin{smallmatrix} 3 \\ 3 \end{smallmatrix}\right]$).

Definition: singular

A square matrix with $\det = 0$ is **singular**. With $\det \neq 0$ it is **nonsingular**.

Notebook `U2-5_Determinants-1_AsArea`, §4, $\det = 0$ checked three ways

Read the Outcome Before Solving

For $ax + by = e, \quad cx + dy = f$: look at $\det \left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$

- $\det \neq 0$: the lines cross. **Exactly one solution**, for **every** right-hand side.
- $\det = 0$: same slope, since $\frac{a}{b} = \frac{c}{d}$ means $ad = bc$. **None or infinitely many.**
- $\det$ sees only the coefficients, so it cannot tell "none" from "infinitely many". Elimination can.

Question 1, answered: the pharmacy

$$x + y = 500, \quad 10x + 2y = 2000 \qquad \Rightarrow \qquad \det \begin{bmatrix} 1 & 1 \\ 10 & 2 \end{bmatrix} = 2 - 10 = -8 \neq 0$$

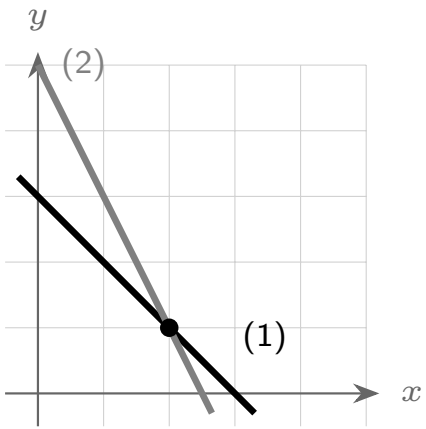
The left side never changes from order to order, so **every** order has exactly one answer.

Next slide: three systems, plotted, with their determinants.

Three Systems, Plotted

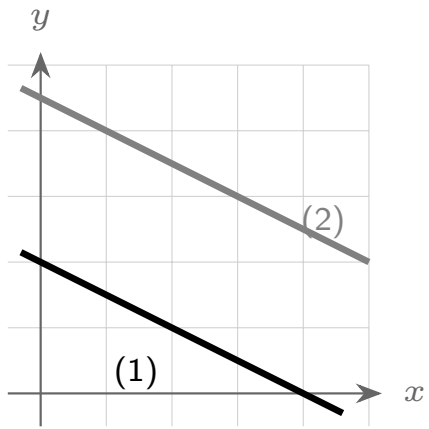
Each line is labelled with its equation's number.

$$x + y = 3$$
$$2x + y = 5$$
$$\det \left[\begin{smallmatrix} 1 & 1 \\ 2 & 1 \end{smallmatrix}\right] = 1 - 2 = -1$$



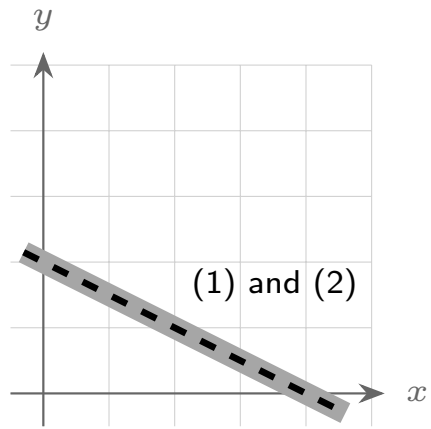
$\det \neq 0$: they **cross**.
Exactly one: $(2, 1)$.

$$2x + 4y = 8$$
$$x + 2y = 9$$
$$\det \left[\begin{smallmatrix} 2 & 4 \\ 1 & 2 \end{smallmatrix}\right] = 4 - 4 = 0$$



$\det = 0$: **parallel**.
No solution.

$$2x + 4y = 8$$
$$x + 2y = 4$$
$$\det \left[\begin{smallmatrix} 2 & 4 \\ 1 & 2 \end{smallmatrix}\right] = 4 - 4 = 0$$



$\det = 0$: the **same line**.
Infinitely many.

Systems 2 and 3 share a coefficient matrix, so they share $\det = 0$. Only the right-hand side tells them apart.

Exercise: Zero or Not?

Try it

1. Without solving, decide whether each system has exactly one solution. If it does, find it; if not, say whether it has none or infinitely many.
   (a) $3x - y = 1, \quad 6x - 2y = 4$ (b) $x + 2y = 5, \quad 3x - y = 1$
   (c) $0.5x + y = 2, \quad x + 2y = 4$
2. The columns of a 3×3 matrix are $\bar{u}$, $\bar{v}$, and $\bar{u} + \bar{v}$. What is its determinant, and why?

Exercise: Solution

Answers

1. (a) $\det = 3(-2) - (-1)(6) = -6 + 6 = 0$: not exactly one. The left side of equation 2 is twice equation 1's, but $4 \neq 2 \times 1$: parallel lines, **no solution**.

   (b) $\det = 1(-1) - 2(3) = -7 \neq 0$: **exactly one**. Elimination gives $(x, y) = (1, 2)$. Check: $1 + 4 = 5$ and $3 - 2 = 1$.

   (c) $\det = 0.5 \cdot 2 - 1 \cdot 1 = 0$. Equation 2 is exactly twice equation 1, right side included: the same line, **infinitely many**.

2. $\det = 0$. The third column is a combination of the first two, so it lies in their plane. All three columns span only a plane, the slanted box is flat, and its volume is 0.

The Inverse

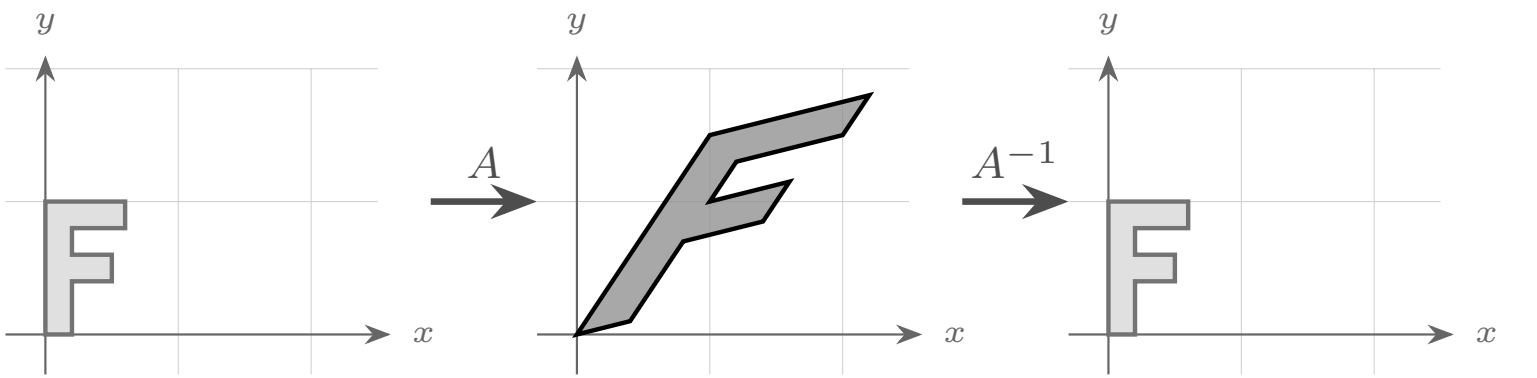
Undoing a Transformation

Definition: the inverse

The **inverse** of a square matrix A is the matrix A^{-1} with

$$A^{-1}A = I \qquad \text{and} \qquad AA^{-1} = I.$$

Apply A, then A^{-1}, and every point is back where it started. A matrix with an inverse is **invertible**; one without is **singular**.



Here $A = \begin{bmatrix} 2 & 1 \\ 0.5 & 1.5 \end{bmatrix}$. Only a **square** matrix can have an inverse.

Notebook `U2-5_Determinants-2_InverseAsUndo`, §1, there and back again

Inverses You Already Know (Unit 2-4)

Move	**Matrix**	**How to undo it**	**Inverse**
Scaling	$\begin{bmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$	halve x, double y	$\begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 2 \end{bmatrix}$
Shear	$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$	slide back the other way	$\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$
Rotation	$R(\theta)$	turn back	$R(-\theta)$
Reflection	F	flip again	F itself
Projection	$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$	**impossible**	none

Check one

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Why projection cannot be undone

$(3, 5)$, $(3, -2)$, and $(3, 0)$ all land on $(3, 0)$. Nothing can know which one to send it back to.

Numbers do the same: $5 \cdot \frac{1}{5} = 1$, and 0 has no reciprocal. For matrices, the role of 0 is played by **every** matrix with $\det = 0$.

The 2×2 Inverse Formula

The formula

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{\det A} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Swap the diagonal, **negate** the off-diagonal, **divide** by $\det A$.

Step 1: swap and negate. This **almost** undoes A

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} ad - bc & -ab + ba \\ cd - dc & -cb + da \end{bmatrix} = \begin{bmatrix} \det A & 0 \\ 0 & \det A \end{bmatrix} = (\det A)\, I$$

Back to I, except everything is still scaled by the factor $\det A$.

Step 2: divide by $\det A$. This removes the leftover scaling

- A multiplies every area by $\det A$, so the undo must multiply every area by $1/\det A$. Indeed $\det(A^{-1}) = 1/\det A$.
- If $\det A = 0$ there is nothing to divide by: a shape squashed to area 0 cannot be scaled back up.

Why the Determinant Sits in the Denominator

At zero: $\det A = 0$ makes the formula divide by zero. A **is invertible exactly when** $\det A \neq 0$**.**

Near zero: $M = \left[\begin{smallmatrix} 1 & 1 \\ 1 & 1+\varepsilon \end{smallmatrix}\right]$, with $\det M = \varepsilon$

$\varepsilon = \det M$	M	M^{-1}
1	$\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$	$\begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$
0.1	$\begin{bmatrix} 1 & 1 \\ 1 & 1.1 \end{bmatrix}$	$\begin{bmatrix} 11 & -10 \\ -10 & 10 \end{bmatrix}$
0.01	$\begin{bmatrix} 1 & 1 \\ 1 & 1.01 \end{bmatrix}$	$\begin{bmatrix} 101 & -100 \\ -100 & 100 \end{bmatrix}$

- M barely changes; M^{-1} **explodes**.
- **Why data science cares:** nearly dependent (collinear) features do this, so a tiny change in the data makes regression coefficients jump.

Notebook `U2-5_Determinants-2_InverseAsUndo`, §2

Worked: a 2×2 Inverse

$A = \left[\begin{smallmatrix} 3 & 1 \\ 1 & 2 \end{smallmatrix}\right]$

1. **Determinant:** $\det A = 3 \cdot 2 - 1 \cdot 1 = 5 \neq 0$, so the inverse exists.
2. **Swap, negate, divide:**

$$A^{-1} = \frac{1}{5} \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix}$$

3. **Check:**

$$AA^{-1} = \frac{1}{5} \begin{bmatrix} 3(2)+1(-1) & 3(-1)+1(3) \\ 1(2)+2(-1) & 1(-1)+2(3) \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} = I$$

The area bookkeeping

A multiplies area by 5, and $\det(A^{-1}) = \frac{1}{25}(2 \cdot 3 - (-1)(-1)) = \frac{5}{25} = \frac{1}{5}$: the undo divides it back by 5.

Notebook `U2-5_Determinants-2_InverseAsUndo`, §2, the formula built by hand

Rules Worth Knowing

Determinants

- $\det(AB) = \det A \cdot \det B$
- $\det(A^{-1}) = 1/\det A$
- $\det(A^\top) = \det A$
- $\det(kA) = k^n \det A \quad (n \times n)$

Inverses

- $(A^{-1})^{-1} = A$
- $(AB)^{-1} = B^{-1}A^{-1}$: **reverse order**
- $(A^\top)^{-1} = (A^{-1})^\top$
- $(kA)^{-1} = \frac{1}{k}A^{-1} \quad (k \neq 0)$

Area factors multiply; the undo divides the area back; kA scales **all** n directions by k, so k^2 for 2×2 and k^3 for 3×3.

Socks and shoes: why the order reverses

Socks on, then shoes on. To undo it, shoes off **first**. Check:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I$$

Watch out: $\det(A + B) \neq \det A + \det B$. Example: $\det(I + I) = \det(2I) = 4$, but $\det I + \det I = 2$.

Exercise: Inverses

Try it

1. Find the inverse of $\begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix}$ and check it by multiplying.
2. Use the formula to invert the quarter turn $R(90°) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Which rotation is the answer?
3. Which of these have an inverse? For any that does, find it.

$$\begin{bmatrix} 4 & 6 \\ 2 & 3 \end{bmatrix}, \qquad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \qquad \begin{bmatrix} 3 & 4 \\ 1 & 3 \end{bmatrix}.$$

4. A and B are 2×2 with $\det A = 3$ and $\det B = -2$. Find $\det(AB)$, $\det(A^{-1})$, and $\det(2A)$.

Exercise: Solution

Answers

1. $\det = 2 \cdot 3 - 5 \cdot 1 = 1$, so swap, negate, divide by 1: $\left[\begin{smallmatrix} 3 & -5 \\ -1 & 2 \end{smallmatrix}\right]$. Check: $\left[\begin{smallmatrix} 2 & 5 \\ 1 & 3 \end{smallmatrix}\right]\left[\begin{smallmatrix} 3 & -5 \\ -1 & 2 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 6-5 & -10+10 \\ 3-3 & -5+6 \end{smallmatrix}\right] = I$. A determinant of 1 is why there are no fractions.

2. $\det = 0 \cdot 0 - (-1)(1) = 1$. Swap the diagonal (two zeros), negate the off-diagonal: $\left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right]$. That is $R(-90°)$, since $\cos(-90°) = 0$ and $\sin(-90°) = -1$: a quarter turn **clockwise** undoes a quarter turn counterclockwise.

3. Check each determinant first:
   - $4 \cdot 3 - 6 \cdot 2 = 0$: **singular**. Column 2 is $1.5\times$ column 1.
   - $0 \cdot 0 - 1 \cdot 1 = -1$: invertible. The formula gives $\frac{1}{-1}\left[\begin{smallmatrix} 0 & -1 \\ -1 & 0 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$: the reflection across $y = x$ is **its own inverse**.
   - $3 \cdot 3 - 4 \cdot 1 = 5$: invertible, with inverse $\frac{1}{5}\left[\begin{smallmatrix} 3 & -4 \\ -1 & 3 \end{smallmatrix}\right]$.

4. $\det(AB) = 3 \cdot (-2) = -6$. $\det(A^{-1}) = \frac{1}{3}$. $\det(2A) = 2^2 \cdot 3 = 12$.

Solving Systems with the Inverse

Solving $A\bar{x} = \bar{b}$ in One Move

Multiply both sides on the left by A^{-1}

$$\begin{aligned} A\bar{x} &= \bar{b} \\ A^{-1}A\bar{x} &= A^{-1}\bar{b} \\ I\bar{x} &= A^{-1}\bar{b} \\ \bar{x} &= A^{-1}\bar{b} \end{aligned}$$

Watch out

- On the **left**, both sides. $\bar{b}A^{-1}$ is not even defined: 2×1 times 2×2.
- For numbers, $3x = 12$ gives $x = \frac{1}{3} \cdot 12$. Matrices have no division; multiplying by A^{-1} takes its place.

Worked: Unit 2-1's system $x + y = 3, \quad 2x + y = 5$

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}, \qquad A^{-1} = \frac{1}{-1}\begin{bmatrix} 1 & -1 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix}$$

$$\bar{x} = A^{-1}\bar{b} = \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix}\begin{bmatrix} 3 \\ 5 \end{bmatrix} = \begin{bmatrix} -3+5 \\ 6-5 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

The same $(2, 1)$ that Gaussian elimination found.

Back to the Pharmacy

Step 1: invert the coefficient matrix, once

$$A = \begin{bmatrix} 1 & 1 \\ 10 & 2 \end{bmatrix}, \quad \det A = 2 - 10 = -8, \quad A^{-1} = \frac{1}{-8}\begin{bmatrix} 2 & -1 \\ -10 & 1 \end{bmatrix} = \begin{bmatrix} -\frac{1}{4} & \frac{1}{8} \\ \frac{5}{4} & -\frac{1}{8} \end{bmatrix}$$

Step 2: order 1, 500 mL at 4 mg/mL

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -\frac{1}{4} & \frac{1}{8} \\ \frac{5}{4} & -\frac{1}{8} \end{bmatrix} \begin{bmatrix} 500 \\ 2000 \end{bmatrix} = \begin{bmatrix} -125 + 250 \\ 625 - 250 \end{bmatrix} = \begin{bmatrix} 125 \\ 375 \end{bmatrix}$$

- Draw 125 mL of the strong stock and 375 mL of the weak one.
- Check: $125 + 375 = 500$ mL, and $10(125) + 2(375) = 1250 + 750 = 2000$ mg.

The same answer as Unit 2-1. The difference shows up on the next order.

Notebook `U2-5_Determinants-2_InverseAsUndo`, §3, solving a system with the inverse

Solve Once, Reuse for Every Order

Order	$\bar{b}$	$A^{-1}\bar{b}$	**Check (mg)**
300 mL at 6	$\begin{bmatrix} 300 \\ 1800 \end{bmatrix}$	$\begin{bmatrix} -75+225 \\ 375-225 \end{bmatrix} = \begin{bmatrix} 150 \\ 150 \end{bmatrix}$	$1500 + 300 = 1800$
1000 mL at 3	$\begin{bmatrix} 1000 \\ 3000 \end{bmatrix}$	$\begin{bmatrix} -250+375 \\ 1250-375 \end{bmatrix} = \begin{bmatrix} 125 \\ 875 \end{bmatrix}$	$1250 + 1750 = 3000$

Every order at once: V mL at c mg/mL, so $\bar{b} = \begin{bmatrix} V \\ cV \end{bmatrix}$

$$x = -\frac{V}{4} + \frac{cV}{8} = \frac{V(c-2)}{8}, \qquad y = \frac{5V}{4} - \frac{cV}{8} = \frac{V(10-c)}{8}$$

- Both amounts are ≥ 0 only when $2 \leq c \leq 10$.
- You cannot mix anything stronger than the strongest stock, or weaker than the weakest.
- One inverse gave a **formula for every order**. Elimination on one order never shows you that.

Determinants and Inverses in Python

NumPy's `linalg` module

```
import numpy as np
A = np.array([[3, 1],
              [1, 2]])
np.linalg.det(A)
# 5.000000000000001
np.linalg.inv(A)
# [[ 0.4 -0.2]
#  [-0.2  0.6]]
b = np.array([5, 5])
np.linalg.solve(A, b)
# [1. 2.]

M = np.array([[2, 1],
              [2, 1]])
np.linalg.inv(M)
# LinAlgError: Singular matrix
```

Think with `inv`, compute with `solve`

- `solve` runs elimination directly on $A\bar{x} = \bar{b}$.
- It is faster, and loses less to rounding, than building `inv(A)` and multiplying.

Watch out

The trailing `...001` is floating point (Unit 1-5). Test whether `det` is **close** to 0, never equal.

Notebook `U2-5_Determinants-2_InverseAsUndo`, §3 and §4

Exercise: Solve with the Inverse

Try it

1. Using the inverse $\begin{bmatrix} 3 & -5 \\ -1 & 2 \end{bmatrix}$ from the last exercise, solve $2x + 5y = 1$, $x + 3y = 2$. Then solve the same left side with right-hand side $\begin{bmatrix} 4 \\ 2 \end{bmatrix}$.
2. The pharmacy is asked for 400 mL at 12 mg/mL. What does the formula $x = \frac{V(c-2)}{8}$, $y = \frac{V(10-c)}{8}$ give, and what does that mean?

Exercise: Solution

Answers

1. $\bar{x} = \begin{bmatrix} 3 & -5 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3-10 \\ -1+4 \end{bmatrix} = \begin{bmatrix} -7 \\ 3 \end{bmatrix}$. Check: $2(-7)+5(3) = 1$ and $-7+3(3) = 2$. With $\begin{bmatrix} 4 \\ 2 \end{bmatrix}$: $\begin{bmatrix} 12-10 \\ -4+4 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$. Check: $4+0 = 4$, $2+0 = 2$. One inverse, two systems.
2. $x = \frac{400(10)}{8} = 500$ mL and $y = \frac{400(-2)}{8} = -100$ mL. A **negative** volume: the order is impossible, because 12 mg/mL is stronger than the strongest stock. The algebra is fine; the request is not.

A First Look at Eigenvectors

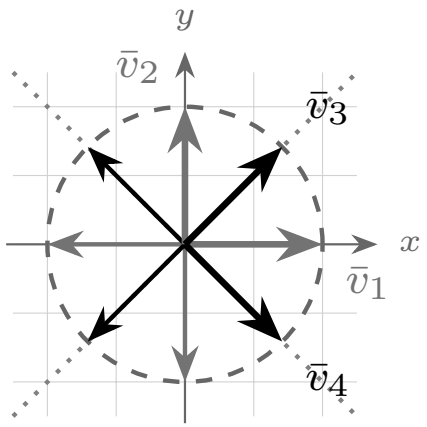
Directions That Do Not Turn

A matrix **turns** most vectors. Are there directions it **only stretches**?

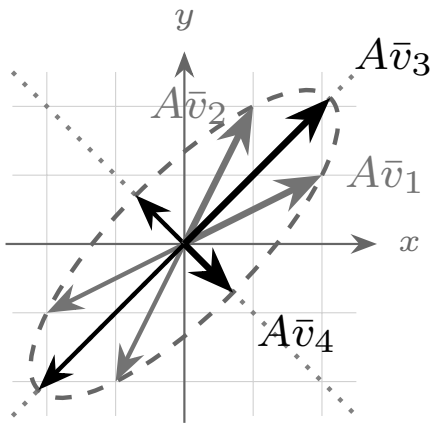
$$A = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}$$

	$\bar{v}$	$A\bar{v}$	**direction**
$\bar{v}_1$	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 0.5 \end{bmatrix}$	turned
$\bar{v}_2$	$\begin{bmatrix} 0 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 0.5 \\ 1 \end{bmatrix}$	turned
$\bar{v}_3$	$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$	$\frac{1.5}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$	**same**, $\times 1.5$
$\bar{v}_4$	$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ -1 \end{bmatrix}$	$\frac{0.5}{\sqrt{2}}\begin{bmatrix} 1 \\ -1 \end{bmatrix}$	**same**, $\times 0.5$

Only $A\bar{v}_3$ and $A\bar{v}_4$ stay on the dotted lines; their negatives do too.



Inputs $\pm\bar{v}$, on the unit circle



Outputs $\pm A\bar{v}$, on the ellipse

Notebook `U2-6_Eigen-1_DirectionsThatDontTurn`, §1, most directions turn

Eigenvectors and Eigenvalues

Definition: eigenvector and eigenvalue

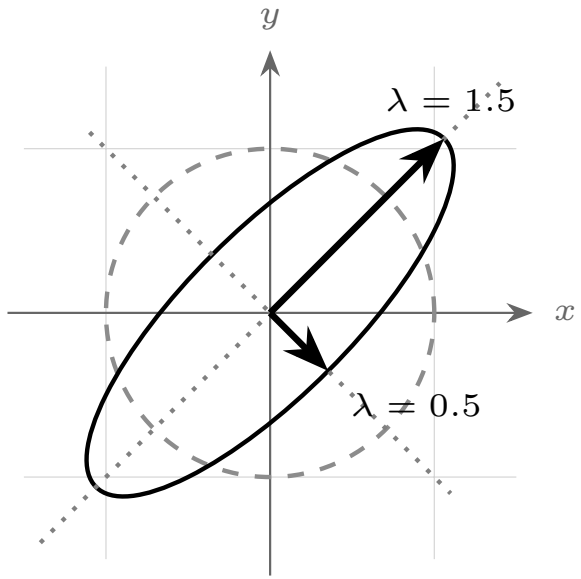
A **nonzero** vector $\bar{v}$ with

$$A\bar{v} = \lambda\bar{v}$$

is an **eigenvector** of A: it is only stretched, never turned. The factor λ is its **eigenvalue**. (*Eigen* is German for "own": A's own directions.)

From the previous slide

- $\bar{v}_3 = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix}$: $A\bar{v}_3 = 1.5\,\bar{v}_3$, so $\lambda = 1.5$.
- $\bar{v}_4 = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\end{bmatrix}$: $A\bar{v}_4 = 0.5\,\bar{v}_4$, so $\lambda = 0.5$.
- $\bar{v}_1$, $\bar{v}_2$: turned, so **not** eigenvectors.



Dashed: the unit circle. Solid: its image under A. The eigenvectors lie along the ellipse's axes.

Notebook `U2-6_Eigen-1_DirectionsThatDontTurn`, §3, the unit-circle animation

Finding the Eigenvalues: $\det(A - \lambda I) = 0$

The determinant does the work

$$A\bar{v} = \lambda\bar{v} \implies A\bar{v} - \lambda I\bar{v} = \bar{0} \implies (A - \lambda I)\,\bar{v} = \bar{0}$$

- $A - \lambda I$ sends a **nonzero** vector to $\bar{0}$.
- An invertible matrix cannot do that: multiplying by its inverse would give $\bar{v} = \bar{0}$.
- So $A - \lambda I$ is **singular**: $\det(A - \lambda I) = 0$, the **characteristic equation**.

Worked: $A = \left[\begin{smallmatrix} 4 & 1 \\ 2 & 3 \end{smallmatrix}\right]$

$$\det(A - \lambda I) = \det\begin{bmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{bmatrix} = (4-\lambda)(3-\lambda) - 1 \cdot 2$$

$$= \lambda^2 - 7\lambda + 10 = (\lambda - 5)(\lambda - 2) = 0$$

So the eigenvalues are $\lambda_1 = 5$ and $\lambda_2 = 2$.

Notebook `U2-6_Eigen-2_CharacteristicEquation`, §1

Why We Care About Eigenvectors

In data science

- **PCA.** Eigenvectors of the covariance matrix are the directions of **greatest spread**. Keep the top few to compress thousands of features into a handful.
- **PageRank.** A page's importance is its entry in an eigenvector of the link matrix.
- **Network communities.** Eigenvectors split a network into groups.

In medicine and science

- **Epidemics.** R_0 is the largest eigenvalue of the next-generation matrix; above 1, an outbreak grows.
- **Diffusion MRI.** The top eigenvector at each point gives the nerve fiber direction.
- **Populations.** The largest eigenvalue is the long-run growth rate.
- **Engineering.** A bridge's vibration modes are eigenvectors.

Common thread: apply a matrix again and again, and the **largest**-eigenvalue direction takes over.

How Many Eigenvalues?

The degree of the polynomial

- For an $n \times n$ matrix, $\det(A - \lambda I)$ is a polynomial of **degree** n: a quadratic for 2×2, a cubic for 3×3.
- So A has **at most** n eigenvalues: exactly n if you count repeats and complex ones.

The rank of A

- **Rank**: the number of linearly independent columns (Unit 2-2).
- $\lambda = 0$ is an eigenvalue $\iff$ $\det A = 0 \iff$ rank $< n$.
- **Symmetric** A: rank = the number of **nonzero** eigenvalues.

A	$\det(A - \lambda I)$	**eigenvalues**	**rank**
$\left[\begin{smallmatrix} 4 & 1 \\ 2 & 3 \end{smallmatrix}\right]$	$\lambda^2 - 7\lambda + 10$	$5,\ 2$	2
$\left[\begin{smallmatrix} 2 & 1 \\ 2 & 1 \end{smallmatrix}\right]$	$\lambda^2 - 3\lambda$	$3,\ 0$	1
$\left[\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right]$	$\lambda^2 - \lambda$	$1,\ 0$	1
$\left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$	$\lambda^2 + 1$	none real	2

Watch out: the last rule needs symmetry. $\left[\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right]$ has eigenvalues $0, 0$ but rank 1.

Finding the Eigenvectors: $A = \left[\begin{smallmatrix} 4 & 1 \\ 2 & 3 \end{smallmatrix}\right]$

From the previous slide, $\lambda_1 = 5$ and $\lambda_2 = 2$. For each one, solve $(A - \lambda I)\,\bar{v} = \bar{0}$. Its two rows say the same thing, so use one.

$\lambda = 5$

$A - 5I = \left[\begin{smallmatrix} -1 & 1 \\ 2 & -2 \end{smallmatrix}\right]$.
Row 1: $-v_1 + v_2 = 0$, so $v_1 = v_2$:
$\bar{v}_1 = \left[\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right]$.

$\lambda = 2$

$A - 2I = \left[\begin{smallmatrix} 2 & 1 \\ 2 & 1 \end{smallmatrix}\right]$.
Row 1: $2v_1 + v_2 = 0$, so $v_2 = -2v_1$:
$\bar{v}_2 = \left[\begin{smallmatrix} 1 \\ -2 \end{smallmatrix}\right]$.

Check

$A\left[\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 4+1 \\ 2+3 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 5 \\ 5 \end{smallmatrix}\right] = 5\left[\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right]$ and $A\left[\begin{smallmatrix} 1 \\ -2 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 4-2 \\ 2-6 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 2 \\ -4 \end{smallmatrix}\right] = 2\left[\begin{smallmatrix} 1 \\ -2 \end{smallmatrix}\right]$.
Eigenvectors are **directions**: any nonzero multiple, such as $\left[\begin{smallmatrix} 3 \\ 3 \end{smallmatrix}\right]$, works just as well.

Notebook `U2-6_Eigen-2_CharacteristicEquation`, §2

What the Eigenvalues Tell You

Their product is the determinant

The area factor is the product of the two stretches.

- $\left[\begin{smallmatrix} 4 & 1 \\ 2 & 3 \end{smallmatrix}\right]$: $5 \cdot 2 = 10 = 12 - 2$.
- $\left[\begin{smallmatrix} 1 & 0.5 \\ 0.5 & 1 \end{smallmatrix}\right]$: $1.5 \cdot 0.5 = 0.75 = 1 - 0.25$.

$\lambda = 0$ means singular

Some direction is crushed to $\bar{0}$. The projection $\left[\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right]$ has $\lambda = 1$ and $\lambda = 0$.

Symmetric matrices behave well

- Always **real** eigenvalues and **perpendicular** eigenvectors.
- $\left[\begin{smallmatrix} 1 & 0.5 \\ 0.5 & 1 \end{smallmatrix}\right]$: $\left[\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right] \cdot \left[\begin{smallmatrix} 1 \\ -1 \end{smallmatrix}\right] = 0$.
- $\left[\begin{smallmatrix} 4 & 1 \\ 2 & 3 \end{smallmatrix}\right]$ is not symmetric: $\left[\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right] \cdot \left[\begin{smallmatrix} 1 \\ -2 \end{smallmatrix}\right] = -1$.

Where this leads: Google's PageRank is an eigenvector, and PCA, a standard way to compress data, is built on the symmetric case.

Notebook `U2-6_Eigen-1_DirectionsThatDontTurn`, §4, symmetric or not

Review: Determinants and Inverses

What we covered: the determinant

1. $\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$ is the **signed area** of the parallelogram spanned by the columns: the factor by which A scales **every** area. In 3D it scales volume.
2. The **sign** records orientation: negative means flipped, as by a reflection.
3. The dot product measures the parallel part, $\|\bar{u}\|\|\bar{v}\| \cos\theta$; the determinant the perpendicular part, $\|\bar{u}\|\|\bar{v}\| \sin\theta$.
4. $\det = 0$ means three things at once: the plane **collapses** to a line, the columns are **dependent**, and A is **singular**. For a system, $\det \neq 0$ means exactly one solution for every right-hand side.
5. $\det(AB) = \det A \cdot \det B$: area factors multiply.

Review: Determinants and Inverses

What we covered: the inverse, and a first look at eigenvectors

6. A^{-1} **undoes** A: $A^{-1}A = AA^{-1} = I$. It exists exactly when $\det A \neq 0$, and $A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. Near $\det = 0$ the inverse blows up.
7. $(AB)^{-1} = B^{-1}A^{-1}$: undo the last move first.
8. $A\bar{x} = \bar{b}$ solves as $\bar{x} = A^{-1}\bar{b}$: invert once, reuse for every right-hand side. In code, prefer `np.linalg.solve`.
9. An **eigenvector** is only stretched, $A\bar{v} = \lambda\bar{v}$. Find λ from $\det(A - \lambda I) = 0$, then $\bar{v}$ from $(A - \lambda I)\bar{v} = \bar{0}$. The eigenvalues multiply to $\det A$.

Notebooks: `U2-5_Determinants-1`, `-2`; `U2-6_Eigen-1`, `-2`. **Previous:** Unit 2-4, Transformations. **Next:** Unit 3, **Calculus**.